PREET MOHAN TYAGI

AI Engineer | Machine Learning Engineer | Generative AI / LLM Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer building end-to-end AI applications across deep learning, NLP, computer vision, and generative AI. Experienced developing FastAPI backends, REST APIs, and LLM-powered features including prompt-engineered chat assistants and retrieval-augmented (RAG-style) document search. Skilled in training/deploying CNN and RNN models and shipping full-stack apps with authentication, CRUD workflows, and cloud databases (PostgreSQL, Supabase). Seeking an AI / ML / GenAI Engineer role.

TECHNICAL SKILLS

Programming Languages: Python, SQL
AI & Machine Learning: Machine Learning, Deep Learning, Neural Networks, CNN, RNN, Scikit-learn, PyTorch, TensorFlow, NumPy, Pandas
Generative AI: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Groq API
NLP: Tokenization, Text Classification, TF-IDF, Stopword Removal, Stemming, NLTK
Computer Vision & Audio: Face Recognition (dlib), Image Processing, Speaker/Voice Verification (Resemblyzer, Librosa)
Backend & APIs: FastAPI, REST APIs, Authentication, CRUD Operations, API Integration, Modular Architecture
Frontend: React.js, HTML, CSS, JavaScript, Streamlit
Databases: PostgreSQL, Supabase
Tools & Deployment: Git, GitHub, Docker, Jupyter Notebook, Linux
Currently Learning / Familiar With: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, Vector Databases (FAISS, ChromaDB, Pinecone), OpenAI API, Google Gemini API, XGBoost, Transformers, OCR, Object Detection

PROJECTS

AI-First CRM (HCP CRM) | *FastAPI, React.js, PostgreSQL, Groq API, Prompt Engineering, REST APIs*

- Architected a full-stack, AI-first CRM (FastAPI + React.js + PostgreSQL) with a modular backend enabling customer management, document handling, and workflow automation.
- Integrated a Groq-hosted LLM with prompt-engineered conversational flows to power an AI chat assistant that answers CRM queries, retrieves customer records, and triggers actions in natural language.
- Built embeddings-based, RAG-style intelligent document search: generated vector embeddings for uploaded documents and ranked results via cosine similarity to ground LLM responses in company data.
- Designed secure REST APIs with authentication and full CRUD operations across customers, documents, and workflows; delivered an analytics dashboard and rule- and LLM-driven workflow automation.

Presencia — AI-Powered Classroom Attendance System | *Python, Streamlit, face_recognition (dlib), Resemblyzer, Supabase*

- Built a full-stack AI attendance application enabling teachers to mark student presence via face recognition or voice verification, replacing manual roll calls.
- Implemented dual-mode biometric verification: dlib-based facial recognition and voice/speaker verification using Resemblyzer and Librosa audio embeddings.
- Designed role-based teacher/student portals with QR-code class-join links (Segno) and a Supabase backend secured via bcrypt password hashing.

Sentiment Analysis on IMDB Reviews using RNN | *PyTorch, NLTK, Scikit-learn*

- Built an end-to-end NLP pipeline (text cleaning, tokenization, stopword removal, Porter stemming) on 50,000 IMDB reviews and engineered TF-IDF features (5,000 dimensions) for sentiment classification.
- Trained a custom RNN classifier in PyTorch (BCE loss, Adam optimizer) for binary sentiment prediction, evaluated via held-out test accuracy.

CIFAR-10 Image Classification using CNN | *PyTorch, Torchvision*

- Designed and trained a 3-block CNN (Conv–ReLU–MaxPool + fully connected layers) for 10-class image classification, achieving 75.4% test accuracy over 10 epochs using Adam optimization and data normalization.

NovaGen — Patient Risk Prediction using Ensemble Learning | *Scikit-learn, Pandas*

- Built and benchmarked 5 classification models (Logistic Regression, KNN, Random Forest, Gradient Boosting, Voting Classifier) for healthcare risk prediction, prioritizing recall to reduce missed high-risk cases; selected Random Forest as the final model (95.8% recall).

CreditWise — Loan Approval Prediction | *Scikit-learn, Pandas, Seaborn*

- Performed EDA and feature engineering (missing-value imputation, categorical encoding, polynomial features) on loan data, benchmarking Logistic Regression, KNN, and Naive Bayes via precision, recall, and F1-score.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University Expected Graduation: 2027

Bachelor of Technology in Computer Science & Engineering (AI/ML) | CGPA: 7.20 / 10